Active Learning Guide

Simulator: Descriptive Graphs of a Sample.

With this application, developed using R and Shiny, you can analyze a sample from four complementary perspectives:

1. Histogram,

2. Box and Whisker Plot

3. Normal Probability Paper

4. Basic Sample Parameters

The goal is to help you improve your ability to describe a sample by combining familiar graphical representations with others that may be less familiar. The application is designed to help you explore how different forms of representation offer different nuances of information about the data distribution.

The histogram is the most intuitive graph for describing the shape of a distribution, and therefore we present it first. Below it, you will find the box and whisker plot, positioned so you can compare it to the histogram at a glance. By having both graphs side-by-side, you can see that they reflect almost the same information: position, dispersion, symmetry, and outliers. The difference is that the histogram shows the peakedness of the distribution, while the box and whisker plot includes more details about the position and dispersion.

Normal probability paper also allows us to visualize position, dispersion, skewness, and outliers, albeit in a different way, with nuances that we can discern by comparing them with the previous graphs. Finally, the application presents several sample statistics, allowing you to quantify what you have observed visually: mean, median, and mode to describe position; standard deviation and interquartile range for dispersion; and skewness and kurtosis coefficients to describe shape. This numerical component allows you to verify whether your graphical interpretation matches the quantitative information, thus strengthening your ability to analyze real-world samples.